

SMART-PORT

TECHNICAL STACK ARCHITECTURE SPECIFICATION

System Classification: Next-Generation Real-Time AIS Maritime Intelligence, Sonar HUD Scanner & Port Optimization Platform

Repository: <https://github.com/Sathiya2-coder/Smart-port> | **Document Version:** 1.0.0

1. Executive Architecture Overview

Smart-Port is an enterprise-grade, high-performance Single Page Application (SPA) designed for real-time maritime telemetry monitoring, high-speed nautical geospatial cartography, acoustic Sonar HUD scanning, and predictive berth congestion analytics. The platform combines a reactive component architecture with hardware-accelerated canvas rendering and low-latency WebSocket stream processing.

2. Detailed Layer-by-Layer Technical Stack

2.1. Frontend Core & Reactive Framework

- **React 18.2.0:** Component-driven declarative UI utilizing concurrent rendering features, functional components, custom hooks, and memoized state hooks (useState, useEffect, useRef, useCallback).
- **Vite 5.1.6:** Next-generation frontend build tool and lightning-fast Hot Module Replacement (HMR) bundler powered by Rollup and native ES Modules (ESM).

2.2. Marine Cartography & GIS Engine

- **Leaflet 1.9.4:** High-performance mapping engine utilizing Hardware-Accelerated Canvas Rendering (preferCanvas: true) for smooth rendering of hundreds of concurrent vessels.
- **CartoDB Dark Matter Tiles:** Ultra-high-contrast nautical vector tile basemaps optimized for night-mode tactical naval operations.
- **Dynamic L.DivIcon & Micro-Badges:** Custom HTML/CSS glowing vessel beacons, true heading vectors, and permanent status pills.
- **Native ResizeObserver Engine:** Real-time container dimension observation ensuring flawless viewport calibration across mobile and desktop displays.

2.3. Motion, Physics & Sonar Radar HUD

- **GSAP 3.15.0 (GreenSock Animation Platform):** Powers 360° continuous polar radar sweeps, pulsing sonar range rings, and interactive micro-animations with 60fps GPU performance.
- **HTML5 Canvas 2D API:** Real-time sinusoidal acoustic waveform generator running on requestAnimationFrame for live sonar telemetry visualization.

2.4. Styling, Design System & Glassmorphism

- **Tailwind CSS 3.4.1 + PostCSS 8.4.35:** Utility-first CSS framework with tailored nautical color palettes and responsive breakpoints.
- **Aesthetic Paradigm:** Glassmorphic HUD with multi-layered backdrop blurs (backdrop-blur-2xl), glowing borders, and monospace telemetry typography.

2.5. Real-Time Data Streaming & Protocol

- **WebSocket Client:** Real-time AIS stream connector (wss://stream.aisstream.io/v0/stream).
- **Zero-Latency Dead-Reckoning Engine:** Mathematical vessel simulator generating live updates at 0ms delay to prevent network dropouts.

2.6. Mathematical & Geodesic Algorithms

- **Haversine Distance Formula:** Exact great-circle trigonometric formula computing true nautical distances (in km & nautical miles) between vessels and ports.
- **True Polar Bearing Projection:** Azimuth angle calculations mapping spherical geographic coordinates onto circular 2D polar radar coordinates ($x = r \cdot \sin \theta$, $y = -r \cdot \cos \theta$).

3. Technology Summary Matrix

System Layer	Core Technology	Version	Primary Role & Capability
Core Framework	React	18.2.0	SPA architecture, reactive state management & component lifecycle
Build & Bundler	Vite	5.1.6	Lightning-fast HMR dev server & optimized production Rollup bundling
GIS Cartography	Leaflet	1.9.4	Interactive marine map, hardware-accelerated canvas vessel rendering
Motion & Radar	GSAP + Canvas	3.15.0	Polar radar sweeps, acoustic wave HUD & micro-interactions
Styling & Tokens	Tailwind CSS	3.4.1	Dark-mode glassmorphism, responsive grid & tactical HUD theme
Iconography	Lucide React	0.344.0	High-clarity nautical, navigation & telemetry SVG icons
Data Pipeline	WebSockets + AIS	Native	Real-time telemetry streaming + zero-delay dead-reckoning engine
Geodesic Math	Haversine / Polar	Algorithm	Accurate spherical distance & polar radar angle projections